



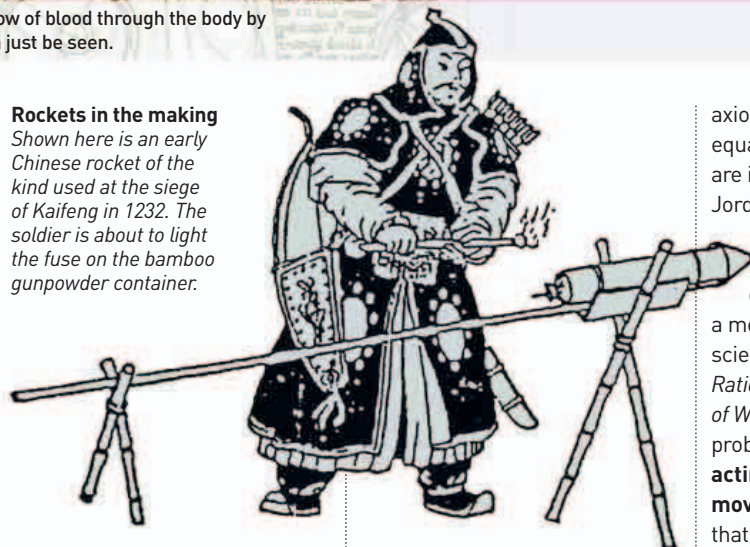
A 13th-century illustration shows the flow of blood through the body by means of veins. The top of the heart can just be seen.

90 THOUSAND MILES

THE LENGTH OF THE ENTIRE NETWORK OF BLOOD VESSELS IN THE HUMAN BODY

EARLY CHINESE GUNPOWDER WEAPONRY had been relatively low powered, in the form of hand-hurled grenades or fire-arrows, with a small charge attached to the shaft. In 1231, faced with a Mongol invasion, the Chinese defenders of Ho Chung deployed the “Heaven-Shaking Thunder Crash Bomb,” which contained **gunpowder rich enough in saltpeter to burst an iron casing**. The resulting explosion could be heard 31 miles (50km) away, and the shrapnel from the explosion was said to have torn iron armor to pieces. In 1232, the **Chinese also used an early form of rocket** consisting of a spear with a bamboo container packed with gunpowder attached to it. When lit, these “fire-spears” were propelled forward by the

Rockets in the making Shown here is an early Chinese rocket of the kind used at the siege of Kaifeng in 1232. The soldier is about to light the fuse on the bamboo gunpowder container.



explosive charge. Another form of the weapon was used as **an early form of flamethrower**, which could shoot fire up to 6.5ft (2m) towards an enemy, causing appalling injuries.

Robert Grosseteste (c.1168–1253), Bishop of Lincoln, played

a key role in **reconciling Aristotelian philosophy and scientific method with Christian thinking** through his commentary on Aristotle’s *Posterior Analytics*, published from 1220 to 1235. His logical method was rigorous, a process he called “**resolution and composition**,” which involved the testing, by experiment if possible, of hypotheses, and the rejection of any conclusions that were not based on observation. His theory that all changes were caused by **the action of forces acting through a medium** led him to study optics and to write treatises on rainbows and astronomy.

Around 1230, European mathematician **Jordanus de Nemore** produced **a new theory of levers** in his *Elementa Super Demonstrationem Ponderum* (*Elements on the Demonstration of Weight*). Building on Aristotle’s

axiom that equal weights at equal distances from a fulcrum are in equilibrium (see pp.34–35), Jordanus introduced the idea of **virtual displacement** (which looks at the effects of infinitesimal changes on a mechanical system) into the science of mechanics. His *De Ratione Ponderis* (*On the Theory of Weight*) also investigated the problem of **downward forces acting along the trajectory of a moving body**. He demonstrated that the more oblique the object’s trajectory, the smaller the downward forces (later understood as positional gravity). Jordanus also developed proof to show the point at which weights supported by angled (or bent) **levers on a fulcrum will be in equilibrium (balanced)**.

Syrian polymath and anatomist **Ibn al-Nafis** (1213–88) produced **a major medical compendium** the *Sharh Tashrīsh al-Qanun* (*Commentary on Anatomy in Ibn Sina’s Canon*). It contained a host of anatomical discoveries, but al-Nafis’s major breakthrough was his discovery of how **blood circulated between the heart and lungs**. He showed that blood circulates from the right-hand side of the heart to the left through the lungs, in contrast to the traditional view of Galen (see 100–250), who held that blood seeped in from the right ventricle



FIBONACCI (c.1170–1250)

Leonardo Pisano (Fibonacci) was born into a wealthy merchant family in Pisa, Italy. His father was in charge of the Pisan trading colony in Bugia in Tunisia, and there Fibonacci came into contact with Arabic mathematical ideas. Aged 32, he published the *Liber Abaci*, which brought him great fame and he gave a mathematical demonstration to King Frederick II of Sicily.

to the left through pores in the wall that separated the two chambers. However, he did not explain how the blood then returned from the heart’s left ventricle to the right. A full theory of blood’s circulation would not be formulated until William Harvey in the 17th century (see 1628–30).

“THE CONSIDERATION OF **LINES, ANGLES, AND FIGURES** IS OF THE GREATEST UTILITY SINCE IT IS IMPOSSIBLE FOR **NATURAL PHILOSOPHY** TO BE KNOWN WITHOUT THEM.”

Robert Grosseteste, English philosopher and theologian, in *On Lines, Angles, and Figures*, c.1235

1220 Robert Grosseteste publishes commentary on Aristotle’s *Posterior Analytics*

c.1230 Jordanus de Nemore develops theory of equilibrium of levers and the idea of virtual displacement

1231 Alf Dakhwar bequeaths money for establishment of first school exclusively for teaching of medicine in Damascus

c.1230–40 Chemical encyclopedia of Bartholomew the Englishman published

1245–46 Syrian physician Ibn Usaybi’ah publishes *Sources of Information on the Classes of Physicians*, first history of Arabic medicine

1247 Song Ci produces first ever work on forensic medicine

c.1248 Islamic physician Ibn al-Baytar compiles the *Comprehensive Book on Simple Drugs*, the most influential medieval Arab pharmacopeia